How accurate LIGHT_Kr v0_5 is

The pure-krypton model LIGHT_Kr v0_5, version 0.5.0-kr, on 16 October 2026: v0_4 with the electrons' collisions with excited krypton and with the excimer taken in, calculated where nothing is measured, each 5s $\rightarrow$ 5p pumping charged its own energy, and the two-temperature closure's tail table given its excited atoms, scored against pure-krypton measurements, with its propagated uncertainty, its reference point and its checks

Bottom line. No measurement in pure krypton covers v0_5's own regime, a gas at 0.1 to 2 atm whose electrons a field heats. The measurements that exist test its parts. Only electron swarm measurements train anything, the cross sections, which v0_5 takes unchanged from v0_3 and v0_4; every other measurement only scores the model.

- **What v0_5 changes.** v0_4 held at 0 the electrons' collisions that ionize the excited atoms, and carried none with the excimer Kr_2^*. Where nothing is measured v0_5 takes a published calculation: the ionization of the 5s levels, of the 5p rung and of Kr_2^* by Vriens and Smeets's formula [69], which for Kr_2^* gives the 10^{-15} cm^2 of Flannery et al.'s calculation [70]. It charges each 5s $\rightarrow$ 5p pumping the energy of the 5p level it reaches (v0_4 charged the gap to the rung's lowest level), returns the 5s levels to the ground state by detailed balance, mixes $1s_5$ and $1s_4$ by electrons at Blagoev and Pilosov's measured rate [71], and gives the two-temperature closure's tail table an axis for the excited atoms.
- **The answer falls by a factor of 5.1.** At LIGHT's reference point (1 atm, 300 kW cm^{-3} for 1 μs) v0_5 sends 6.08 % of the input to 147 nm on the kinetic EEDF; v0_4 sent 31.22 %. The electrons ionize 56.9 % of the excimers before they radiate (28.5 % do), and the stepwise ionizations and the excimer's make most of the ions, which recombine onto the 5p rung, whose relaxation heats the gas. With the three calculated rows at v0_4's 0 the share is 28.02 %.
- **Propagated from its sources**, the share carries $\pm$7.0 points in quadrature ($\pm$115 %), and the trained cross sections' posterior $\pm$0.06 (Section 1). The three largest terms are the electrons' collisions with Kr_2^*: its ionization, a calculation (the share 15.43 % without it, 4.47 % at twice it), and its quenching and the mixing of its two states, which nothing measures or calculates (1.27 and 12.44 % at Kannari et al.'s estimate).
- **A new score.** Blagoev et al. measured the 5s levels' return to the ground state by thermal electrons in an afterglow [68]. v0_5's, the detailed balance of v0_3's trained excitation cross sections, lies at 5.3 to 6.6 times theirs for $1s_5$, 5.5 to 9.5 for $1s_4$ and 0.5 to 1.2 for $1s_3$ (Section 5); the return moves the reference point by -0.5 %.
- **The electroluminescence yield** of Mano et al. [66] and **the electron swarm** are v0_4's: 0.99 to 1.04 of the yield's fit from 6 to 13.5 Td; the drift velocity within 2.8 % rms and the ionization coefficient within 9.9 % on the train sets (Sections 7 and 6).
- **Under an electron beam** the new rows act only where the beam's plasma is dense: v0_5 turns 0.543 of the deposited energy into 147-nm light at low pump, v0_4's value, where Morozov et al. measured 0.42 $\pm$ 0.05 [38] (+29 %), and at 3×10^5 W cm^{-3} and 1 atm it is -3 % from Eckstrom et al.'s report (v0_4: +1 %; Section 8).
- **The two-temperature closure** on its tail table with the excited axis gives 5.58 % at the reference point, -8 % from the kinetic EEDF; on the table without the axis (v0_4's) 6.30 %, +4 %. Its tail now follows the kinetic one, and its Maxwellian bulk is what is left (Section 4).
- **The numerical error is far smaller.** The energy ledger closes to 4×10^{-15} of the input; halving the kinetic closure's renewal steps or doubling its cells moves the 147-nm and near-IR shares by at most 1×10^{-3} of themselves.

Table 1 lists every test, Section 1 reads them as errors and propagates the model's own uncertainty, and the sections after it give each test's detail. The swarm datasets in the role *train* set the cross sections' factors (Section 6); every other measurement in the table only scores the model, and no measurement set any other rate, which come from the sources in Appendix A.

test	measured	v0_4's error	v0_5's error	role
energy ledger closure, reference point			4×10^{-15} of E_{dep}	bookkeeping
energy balance at every node of the flow network			30 nodes, to 10^{-8}	bookkeeping
drift velocity, the 3 sets that agree [40, 42, 33], 79 points		rms 2.8 % (one set of cross sections)		train
ionization coefficient, 50 to 300 Td, the 5 sets that agree [49, 43, 5, 35, 36], 64 points		rms 9.9 %		train
drift velocity, the 8 sets that do not agree (1953 to 1992)		rms 8 to 47 % by set		held out; the two at 195 K validation
characteristic energy D_T/μ, Koizumi et al. [47], 26 points, 0.005 to 0.9 Td		rms 7.1 %		validation
ionization coefficient below 50 Td, 5 sets [50, 35, 36, 43, 5], 36 points		mean -36 %		held out by rule
electroluminescence yield, 6 to 13.5 Td, Mano et al. [66]	$0.113\,E/N - 0.3$	0.99 to 1.04 of it		held out
superelastic return of $1s_5$, $1s_4$, $1s_3$ by thermal electrons, Blagoev et al. [68]	1.5×10^{-10}, 5.2×10^{-10}, 9.8×10^{-10} cm^{3}s^{-1}	not carried	$\times$5.3–6.6, $\times$5.5–9.5, $\times$0.5–1.2	new, held out
ionization coefficient, Monte Carlo on the same cross sections [3], 20 to 50 Td		Biagi's file +3.7, -4.0, -3.5, -2.5 %		solver check
the Monte Carlo's bulk drift velocity and SST coefficient, trained set against Biagi's file, 30 to 300 Td		within 1.6 %		solver check
energy per ion pair under a beam, from Saito et al. [37]	23.9 eV	+10 %		held out
excitations per ion pair, Saito et al. [37]	0.55	+24 %		held out
energy per excitation or ion, Saito et al. [37]	15.4 eV	+1.0 %		the beam's check
VUV efficiency, continuous e-beam, Morozov et al. 2008 [38]	0.42 ± 0.05	+29 %	+29 %	held out
VUV efficiency at 3×10^5 W cm^{-3}, Eckstrom et al. 1988 [39]	near 0.46, their limit	+1 %	-3 % at 1 atm	held out

Table 1: LIGHT_Kr v0_5 against the data, every band at its default, on the kinetic EEDF, beside v0_4's errors. v0_5 runs on v0_3's and v0_4's trained cross sections, so the swarm and the beam's partition score one set of cross sections. The superelastic row gives v0_5's rate over the measurement, for electrons at 300 to 2000 K. An error is the model less the measurement over the measurement, signed for a single value and rms (or a mean) over a set. A validation set is held out of the training but took part in choosing which factors train (Table 10). The swarm rows compare the model as each experiment measures it: the two-term solution times the Monte Carlo's ratios (Section 6). The beam rows run under the e-beam drive of Section 8.

1 Overall accuracy and its uncertainty

Against the data. Five groups of measurements bear on v0_5, and each tests a part of it (Table 1).

- **The electron swarm** (Section 6) trains the cross sections that give every electron rate and scores what the training did not see. v0_5 runs on v0_3's trained set. On the 8 sets that agree with one another the drift velocity is within 2.8 % rms and the ionization coefficient within 9.9 % from 50 to 300 Td. Held out, the characteristic energy of Koizumi et al. is within 7.1 % rms (Biagi's file: 15.8 %), a validation rather than a test, as it took part in choosing which factors train. The ionization coefficient below 50 Td, which nothing in the training saw, lies -36 % from the measurements on average, where

those part by up to $\times 4.0$ from one another.

- **The electroluminescence yield** (Section 7), a swarm measurement of the light the excitation makes, which no fit saw: the model lies at 0.99 to 1.04 of it from 6 to 13.5 Td.
- **The superelastic return** (Section 5) is a new score. v0_5 returns the 5s levels to the ground state by detailed balance on the trained excitation cross sections; for thermal electrons that gives 5.3 to 6.6 times Blagoev et al.'s afterglow rate for $1s_5$, 5.5 to 9.5 for $1s_4$ and 0.5 to 1.2 for $1s_3$.
- **The e-beam measurements** (Section 8) score the chain from excitation to the 147-nm photon. The beam's partition is v0_4's: +10 % from Saito et al. on the energy per ion pair and +24 % on the excitations per pair. At low pump the error against Morozov et al. is +29 %, 2.5 of their standard errors. At 3×10^5 W cm^{-3} and 1 atm it is -3 % against Eckstrom et al.'s report.
- **Nothing scores the field-driven regime.** No pure-krypton measurement in hand tests a gas whose electrons a field heats at the reference point's density. There v0_5's answer rests on the electrons' collisions with excited atoms and with Kr_2^*, which v0_4 held at 0: their rates are a calculation that no krypton measurement tests, and their band is propagated below.

Propagated from the sources. Each input is set to the two ends its source allows and the run repeated, the others at their defaults, on both electron closures (Table 2). The inputs are the 21 bands the rows read and the 3 measured constants with a quoted error that no band covers: $Kr_2^+ + e^-$ (± 10 % [30]), $Kr^+ + 2Kr$ (the spread of the three measurements [26, 27, 28]) and the Kr_2^* lifetime (270 ± 5 ns [25]). A calculated row's band runs from 0 to twice the calculation. Half the difference between an input's two ends is its term. The trained cross sections are one more input: the reference point is run at 12 draws of the training's posterior on the kinetic EEDF and 40 on the two-temperature closure, and the draws' standard deviation is their term. The quadrature sum treats the terms as independent standard deviations, which a band's range is not, so it is indicative. The excursions from the reference, summed by sign, are the ceiling of the spread, not a bound. Not propagated: the 5p rung's lines and its levels' population at the gas temperature (its light is under 1 % of the input), and the excimer's rotational constant, an estimate that scales its redissociation as $1/r_e^2$ while the well's band moves it exponentially.

On the 147-nm share at the reference point, 6.08 % of the input on the kinetic EEDF, the quadrature sum is ± 7.0 points (± 115 %). The excursions sum to -13.2 and +21.5 points. The largest terms are the ionization of Kr_2^* by electrons (5.48 points), the electrons' mixing of the excimer's states (3.18) and the quenching of Kr_2^* by electrons (2.41). The trained cross sections' term is 0.06 points (1.0 % of the share). On the two-temperature closure the sum is ± 7.2 points (± 129 %), the cross sections' term 0.14.

What dominates has changed again. In v0_4 it was the $J = 1$ levels' pumping; in v0_5 the three largest terms are the electrons' collisions with Kr_2^*, which v0_4 did not carry, and the first alone makes 61 % of the quadrature sum's square. The excimer's ionization, Vriens and Smeets's calculation, takes the share from 15.43 % without it to 4.47 % at twice the formula. The two rows that nothing measures or calculates move it as far from their 0: at Kannari et al.'s estimate, 10^{-7} cm^{3}s^{-1}, the excimer's quenching by electrons takes it to 1.27 %, and the same rate mixing its two states raises it to 12.44 %, the fast 0_u^+ state radiating in 3.4 ns, before the electrons can ionize it. Of the measured inputs, the $1s_4$ formation's band, down to a third of the $1s_5$ rate (Audouard et al.'s ratio [56]), takes the share to 4.29 %, and the measured 5s $\rightarrow$ 5p cross sections' quoted errors span 5.46 to 7.06 %. The $J = 1$ levels' proxies, v0_4's largest term, now move it the other way and less: with no electron pumping them the share is 5.55 %.

input	its two ends	147 nm, kinetic	147 nm, two-temperature	near IR, kinetic	T_g (K), kinetic
$e^- + Kr_2^* \rightarrow Kr_2^+ + 2e^-$, scale	0 to 2	15.43 to 4.47	15.54 to 4.06	0.43 to 0.49	801 to 866
$e^- + Kr_2^* \rightarrow e^- + Kr_2^*(0_u^+)$	0 to 1.0×10^{-7} cm^3 s^{-1}	6.08 to 12.44	5.58 to 12.08	0.48 to 0.44	856 to 820
$e^- + Kr_2^* \rightarrow e^- + 2Kr$	0 to 1.0×10^{-7} cm^3 s^{-1}	6.08 to 1.27	5.58 to 0.94	0.48 to 0.52	856 to 885
$Kr^*(1s_4) + 2Kr \rightarrow Kr_2^* + Kr$	1.3×10^{-32} to 9.2×10^{-32} cm^6 s^{-1}	4.29 to 6.08	3.89 to 5.58	0.51 to 0.48	866 to 856
$e^- + Kr^* \rightarrow Kr^{**}$, scale	0.6 to 1.4	7.06 to 5.46	6.32 to 5.12	0.45 to 0.50	850 to 860
$e^- + Kr^* \rightarrow Kr^+ + 2e^-$, scale	0 to 2	6.96 to 5.60	6.60 to 5.15	0.49 to 0.47	851 to 859
$Kr^{**} + Kr \rightarrow Kr^* + Kr$	3.0×10^{-11} to 1.7×10^{-10} cm^3 s^{-1}	7.11 to 6.08	6.39 to 5.58	2.11 to 0.48	840 to 856
$Kr_2^* + Kr_2^*$, Penning	0 to 5.0×10^{-10} cm^3 s^{-1}	6.08 to 5.12	5.58 to 4.79	0.48 to 0.49	856 to 862
the $1s_4$ formation's share into 0_u^+	0 to 0.17	6.08 to 6.95	5.58 to 6.51	0.48 to 0.48	856 to 851
$Kr^*(1s_5) + 2Kr \rightarrow Kr_2^* + Kr$	1.7×10^{-32} to 4.2×10^{-32} cm^6 s^{-1}	5.34 to 6.20	4.88 to 5.70	0.49 to 0.48	860 to 855
the excimer's well (its redissociation)	0.62 to 0.709 eV	5.35 to 6.08	4.90 to 5.58	0.49 to 0.48	859 to 856
$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$, scale	0 to 2	6.37 to 5.83	5.86 to 5.36	0.49 to 0.47	854 to 858
$Kr^*(1s_3) + 2Kr \rightarrow Kr_2^* + Kr$	0 to 9.2×10^{-32} cm^6 s^{-1}	5.81 to 6.34	5.33 to 5.83	0.48 to 0.48	857 to 855
$e^- + Kr^*(J = 1) \rightarrow Kr^{**}$, scale	0 to 1	5.55 to 6.08	4.76 to 5.58	0.32 to 0.48	858 to 856
$Kr_2^+ + e^-$, the gas-temperature factor	0 to 1	6.08 to 6.50	5.58 to 5.99	0.48 to 0.47	856 to 854
$Kr^* + Kr^*$, Penning	1.5×10^{-10} to 5.0×10^{-10} cm^3 s^{-1}	6.50 to 6.08	5.98 to 5.58	0.48 to 0.48	853 to 856
its products over the 5s levels, share	0 to 1	6.36 to 6.08	5.89 to 5.58	0.47 to 0.48	855 to 856
$Kr_2^* + Kr \rightarrow 3Kr$	0 to 8.6×10^{-15} cm^3 s^{-1}	6.08 to 5.88	5.58 to 5.41	0.48 to 0.48	856 to 857
$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.2×10^{-31} to 3.3×10^{-31} cm^6 s^{-1}	6.04 to 6.20	5.54 to 5.72	0.48 to 0.48	856 to 855
the electron cross sections (trained)	12 and 40 draws of the posterior	6.11 ± 0.06	5.60 ± 0.14	0.48 ± 0.00	856 ± 0
$Kr_2^+ + e^- \rightarrow Kr^{**} + Kr$	± 10 %	6.02 to 6.13	5.53 to 5.64	0.48 to 0.48	856 to 856
Kr_2^* lifetime	265 to 275 ns	6.03 to 6.13	5.54 to 5.63	0.48 to 0.48	856 to 856
the resonance lines' escape	0 to 1.7×10^5 s^{-1}	6.08 to 6.05	5.58 to 5.55	0.48 to 0.48	856 to 856
$e^- + Kr^*(1s_5) \rightarrow e^- + Kr^*(1s_4)$	4.3×10^{-8} to 2.3×10^{-7} cm^3 s^{-1}	6.08 to 6.10	5.58 to 5.60	0.48 to 0.48	856 to 856
$Kr(nl) + Kr \rightarrow Kr_2^+ + e^-$, the share	0 to 1	6.08 to 6.08	5.58 to 5.58	0.48 to 0.48	856 to 856

Table 2: The propagated uncertainty's inputs, each alone at its two ends with the others at their defaults, and the outputs at those ends (low end to high end), percent of E_{dep}; for the trained cross sections the mean and standard deviation over the posterior's draws. With every input at its default the 147-nm share is 6.08 % (kinetic) and 5.58 % (two-temperature).

Numerical accuracy.

- **The ledger.** The energy ledger closes to 4×10^{-15} of the input on the kinetic EEDF and to 2×10^{-14} on the two-temperature closure. The energy flows also balance at each of 30 nodes to 10^{-8}: the electron gas, each reservoir, each group of rows and each output.
- **The kinetic EEDF at the reference point.** The run takes 338 kinetic solutions, 263 in the pulse, at E/N from 1.34 to 6.4 Td. Over the pulse's second half the field carries 1.0001 to 1.0002 of the input. There the kinetic and the model's energy balances agree to 1×10^{-4} of the input, and the kinetic mean energy equals the model's to 1×10^{-10}.
- **Convergence.** Half the renewal step moves the 147-nm share by 2×10^{-6} of itself. A renewal at least every 5 ns moves it by 5×10^{-5}, and twice the energy cells by 1×10^{-3}. A ten times tighter solver tolerance moves it by 8×10^{-8}, the EEDF renewed 338 times against the reference's 338.
- **The afterglow's EEDF.** Through the pulse the kinetic EEDF is advanced in time from its last solution; after the pulse it takes its steady shape at each renewal. Advancing it in time after the pulse too (75 time steps) moves the 147-nm share by 2×10^{-4} of itself, the near IR by 2×10^{-5} and the reaction heat by 1×10^{-5}: the steady shape costs nothing at this point.
- **Restarts.** Where the kinetic EEDF finds no solution from its warm start, the solver starts it again cold. The mean match has one solution from any start, so a restart changes no answer, only the way to it. Over the 72 kinetic runs of this note the solver restarted 43 times, in two runs. Where a table point told to leave the growth term out has no steady solution, as in the afterglow of very hot gas, where Penning ionization makes electrons faster than recombination takes them, it takes the growth term on the same grid, and failing that a grid of its own: 42 times, in one run. Before v0_4 such a point stopped the run. Where even that finds none, v0_5 leaves the point out of the table (11 times, in one run), and where the table's centre has none, LIGHT's Maxwellian stands in until the next renewal (never); before v0_5 either stopped the run, as one did in the afterglow of the scan's 0.25-atm, 500-kW corner.
- **The swarm scores** are independent of the grid: half and twice the 1000 energy cells move the train sets' rms by 0.01 points at most. The emulator the training ran on reproduces the solver to 0.10 % in the drift velocity and 0.74 % in the ionization coefficient on designs it was not trained on.

2 What v0_5 is

LIGHT_Kr v0_5 is a 0-D model: one uniform box of pure krypton at constant volume, into whose electrons a given power is put. It follows where that energy goes: into the four 5s levels (Kr^*: $1s_5$, $1s_4$, $1s_3$, $1s_2$) and the 5p rung (Kr^{**}), the ions Kr^+ and Kr_2^+, the excimer Kr_2^* in its slow (1_u, 0_u^-) and fast (0_u^+) states and its 147-nm second continuum, the 5p $\rightarrow$ 5s lines, the resonance lines, heat, and the electron gas. Its energy ledger closes by construction. Every species carries a formation energy, and every row's energy goes to the electrons, the gas, light or storage.

Its network has 219 rows on 10 species (Appendix A):

- electron-impact excitation of each of the 51 levels of Biagi's cross sections at its own threshold, ionization and elastic loss [1, 6, 7], with v0_3's factors trained on the swarm data (Section 6): each 5s level makes its own species, every higher level the 5p rung Kr^{**} (the higher levels cascading through it, the excess heating the gas);
- the Hornbeck–Molnar ionization of the 20 levels at or above Kr_2^+'s energy [4], a band at 0;
- Penning ionization of the 5s levels on one another;
- ion conversion $Kr^+ \rightarrow Kr_2^+$ and its reverse, and dissociative recombination of Kr_2^+ onto the 5p rung;
- three-body excimer formation out of each 5s level at its own rate, its reverse, the 147-nm emission of the two excimer states, and their quenching by krypton (a band at 0);
- the transfers between the 5s levels on krypton and their reverses, and the escaping resonance lines (a band at 0);
- the rung's relaxation on krypton and its reverse, and its 28 5p $\rightarrow$ 5s lines, each from its own level, the rung's ten levels populated at the gas temperature;
- the electrons' 5s $\leftrightarrow$ 5p collisions, each of the 26 measured pairs [9, 10] at its own threshold, and their reverses;
- the electrons' collisions with the excited atoms and the excimer: the 5s levels' and the 5p rung's ionization and Kr_2^*'s, on Vriens and Smeets's calculation [69]; the 5s levels' return to the ground state; the mixing of $1s_5$ and $1s_4$; and, as bands at 0, the excimer's quenching and the mixing of its two states;
- and, as bands at 0, the other rows with no measurement.

The electrons' rates come from the two-term Boltzmann equation with electron-electron collisions [8], solved during the run at the model's own mean energy (the default, minutes a run), every collision with an excited population a process of its own in it (74 processes). The alternative is a Maxwellian with a tabulated tail (the two-temperature closure, seconds a run; Section 4). A rate known only as a published spread, or not at all, is a band: a default and two ends.

What changed from v0_4. v0_4 had 126 rows.

- **Calculations where nothing is measured.** The ionization of the four 5s levels and of the 5p rung by electrons, which v0_4 held at 0 (no measurement for krypton or argon), is Vriens and Smeets's cross section [69] at each level's own ionization energy, 3.356 to 4.084 eV for the 5s levels; the ionization of Kr_2^*, which v0_4 did not carry, the same at the excimer's 3.644 eV, the order of Flannery et al.'s binary-encounter calculation [70]. Each is a band from 0 to twice the calculation. The excimer's quenching by electrons, $e + Kr_2^* \rightarrow e + 2Kr$, and the mixing of its two states are neither measured nor calculated in the sources found: bands at 0, to Kannari et al.'s estimate of the quenching [11].
- **The pumps' energy.** v0_4 charged every 5s $\rightarrow$ 5p pumping the gap to the rung's lowest level, $2p_{10}$. v0_5 books each measured pair as its own row, the electron paying the pair's threshold and the level's excess over $2p_{10}$ heating the gas as the rung's levels relax: on average 1.537 eV out of $1s_5$ (v0_4: 1.388), 1.420 out of $1s_4$ (1.271), 1.561 out of $1s_3$ (0.741) and 1.479 out of $1s_2$ (0.660). The rung's ten levels are taken populated at the gas temperature (an assumption, Section 10); its superelastic return to each 5s level and its lines are read from each level, and its relaxation's reverse is detailed balance at the gas temperature.
- **The 5s levels and the electrons.** Each 5s level returns to the ground state by the Klein–Rosseland

reverse of its own excitation cross section; $1s_5$ and $1s_4$ mix by electrons at Blagoev and Pilosov's measured $4.3 \times 10^{-8}\,\mathrm{cm^3 s^{-1}}$ [71] (a band to 2.3×10^{-7}, the electron temperature of their afterglow not being stated in their abstract).

- **The tail table's excited axis.** Section 4.
- **Data.** Kr_2^+ is bound by Signorell et al.'s $9267.8\,\mathrm{cm^{-1}}$ [61]; Kholin et al.'s $1s_3$ measurement [67] has argon for its third body, so $1s_3$'s excimer formation stays a proxy.

The cross sections, the swarm, Jung et al.'s tables and every row v0_5 did not write are v0_4's. Table 3 undoes each change alone at the reference point.

undone from v0_5	147 nm, kinetic	change	two-temperature	T_g end (K)
none (v0_5)	6.08		5.58	856
the three calculated rows at 0 (v0_4's rule)	28.02	+361	30.54	726
no stepwise ionization (5s and 5p)	7.50	+23	7.25	847
no ionization of Kr_2^* by electrons	15.43	+154	15.54	801
no superelastic return of the 5s levels	6.11	+0.5	5.80	856
no $1s_5 \rightarrow 1s_4$ by electrons	6.07	−0.2	5.58	856
no reverse of the 5p rung's relaxation	6.07	−0.1	5.58	856
every level forms the excimer at Ueda's 1.7×10^{-32} (v0_3's)	3.16	−48	2.86	873
no redissociation of Kr_2^*	6.42	+5.5	5.91	855
the $J = 1$ levels pumped by no electron	5.55	−8.6	4.76	858
v0_4	31.22		32.92	707

Table 3: What each of v0_5's changes does at the reference point, and two of v0_4's: each undone alone, the others kept, percent of E_{dep}; change is relative to v0_5 on the kinetic EEDF. v0_4 is the last row. The pumps' energy cannot be undone alone; the text gives its size.

The calculated rows are most of the change. With the three at v0_4's 0 the share is 28.02 %, below v0_4's 31.22 %: the rest of v0_5 lowers it, chiefly the changes that cannot be undone alone, the pumps' energy, each pair charged its own threshold (19.0 % of the input to heat), and the rung's levels at the gas temperature. The excimer's ionization alone moves the share by −61 %: without it the share is 15.43 %. The stepwise ionizations move it by −19 %, the superelastic return by −0.5 %, the electrons' $1s_5 \rightarrow 1s_4$ by +0.2 %, and the rung's reverse by +0.1 % at the reference point's 856 K. With every level forming the excimer at Ueda's rate, v0_3's, the share is 3.16 %, and without the excimer's redissociation 6.42 %. The pumping of the two $J = 1$ levels now raises the share (5.55 % without it), where in v0_4 it lowered it.

3 The reference point

LIGHT's drive in pure krypton: 1 atm at 300 K, $300\,\mathrm{kW\,cm^{-3}}$ for 1 μs into the electrons, a $10^{15}\,\mathrm{cm^{-3}}$ seed, 2 μs in all. Table 4 gives its outputs on both closures.

	kinetic EEDF (default)	two-temperature
147 nm, Kr_2^*	6.08	5.58
near IR, 5p $\rightarrow$ 5s	0.48	0.48
heat, reactions and relaxation	90.69	90.67
heat, elastic collisions	3.23	3.75
stored in ions (the seed's ions recombine)	-0.74	-0.74
to Kr^* and Kr^{**} by electron impact, gross	9.37	8.75
to Kr^+	42.96	47.11
to Kr_2^* (formed $\times$ 8.43 eV)	21.37	21.45
the 5s $\rightarrow$ 5p pumps' excess over $2p_{10}$, to heat (gross)	19.0	18.7
Kr^* formed that becomes Kr_2^* (%)	4.4	4.4
Kr^* formed that the electrons ionize (%)	1.9	2.2
Kr^* formed that returns to the ground state by electrons (%)	0.13	0.14
Kr^{**} formed that the electrons ionize (%)	0.9	0.9
Kr_2^* formed out of $1s_5$, $1s_4$, $1s_3$ (%)	43.9, 50.6, 5.5	43.7, 50.8, 5.5
Kr_2^* formed that radiates (%)	28.5	26.0
Kr_2^* formed that the electrons ionize (%)	56.9	59.9
Kr_2^* formed that redissociates (%)	14.0	13.5
ions made by Kr_2^*'s ionization (% of all, by number)	45.2	43.7
by the 5s levels' ionization	33.8	36.7
by the 5p rung's ionization	16.0	15.2
by Penning ionization	4.9	4.5
from the ground state	0.1	0.0
5s $\rightarrow$ 5p pumpings per Kr_2^* formed, net	25	24
T_e, pulse mean (eV)	1.57	1.53
n_e, pulse mean (cm^{-3})	8.2×10^{14}	8.6×10^{14}
T_g at the end (K)	856	859
n_e at the end (cm^{-3})	1.1×10^{13}	1.1×10^{13}

Table 4: The reference point at t = 2 μs, percent of E_{dep} unless a row says otherwise. The fate rows (the first five, with the other stores) sum to 100; the flows below them overlap.

The electrons pump Kr^* to the 5p rung 25 times for each Kr_2^* formed net, and krypton relaxes the rung back, so the cycle's thresholds, charged pair by pair, become heat: 73.3 % of the input, 102 eV for each 147-nm photon. Of the pumping, 44.4 % is out of $1s_5$, 25.5 % out of $1s_4$, 22.8 % out of $1s_2$ and 7.4 % out of $1s_3$. The excimers that form mostly do not radiate: the electrons ionize 56.9 % of them, 28.5 % radiate and 14.0 % redissociate. Counted by number, the excimer's ionization makes 45.2 % of the ions, the 5s levels' 33.8 %, the rung's 16.0 %, Penning ionization 4.9 % and the ground state 0.1 %, and every ion recombines onto the 5p rung (R11p): an excimer ionized is an atom sent round the cycle again. A stepwise ionization costs the electron a fraction of the ground state's 13.9996 eV, and the pulse holds 8.2×10^{14} cm^{-3} of electrons (v0_4: 2.8×10^{14}) at 1.57 eV (v0_4: 2.55). On the kinetic EEDF the share differs from the two-temperature closure's by +9 %. The cycle's measured cross sections, scaled within their errors and against Kannari's fit in their place [11, 12]:

5s $\rightarrow$ 5p cross sections	0 (no cycle)	$\times$ 0.6	$\times$ 1, Jung et al.	$\times$ 1.4	Kannari's fit
147 nm, kinetic EEDF	11.4	7.1	6.1	5.5	6.7
147 nm, two-temperature	9.8	6.3	5.6	5.1	6.1

Table 5: The 147-nm share (percent of E_{dep}) against the scale on the measured 5s $\rightarrow$ 5p cross sections, every pair's. The band runs from 0.6 to 1.4, the measurements' quoted errors; the two J = 1 levels' proxies have a band of their own (`k_5s5p_j1`, Table 2).

Jung et al. measured the excitation out of the two metastable levels, $1s_5$ and $1s_3$, into the ten 5p levels, with stated uncertainties of $\pm$35 and $\pm$40 % [10]. For the two J = 1 levels nothing is measured; each takes the set of the metastable with the same ion core, shifted to its own threshold (Zeng et al.'s R-matrix calculation covers them [73]; its numbers are not in a source in hand). At the reference point's 1.57 eV the four levels' Maxwellian rates, each the sum of its pairs, are 5.2×10^{-7}, 5.4×10^{-7}, 6.0×10^{-7} and

6.2×10^{-7} cm^{3}s^{-1} ($1s_5$, $1s_4$, $1s_3$, $1s_2$); the pairs' sum is v0_4's level rate to 1×10^{-5} of it (their onsets at Biagi's level energies, where Table I of the measurement prints two decimals).

Pressure and power move the share as much (Table 6, two-temperature closure). It rises with pressure and falls with power at every point of the scan, from nothing at 0.1 atm to 20 % at 2 atm and 100 kW cm^{-3}. At 0.1 atm a microsecond of this drive heats the gas to 1354 to 3064 K, beyond the measured range of several rates, and at 500 kW cm^{-3} 67 % of the input is still in ions at the end.

kW cm^{-3}	0.1 atm	0.25 atm	0.5 atm	1 atm	2 atm
100	0.02	0.90	5.17	12.29	19.85
300	0.00	0.02	0.65	5.58	12.38
500	0.00	0.00	0.15	2.24	9.24

Table 6: The 147-nm share (percent of E_{dep}) over pressure and power, 1 μs pulses, the two-temperature closure on its field-driven tail table.

4 The two-temperature closure and its tail

The two-temperature closure keeps the electrons Maxwellian at T_e below krypton's lowest threshold, 9.9152 eV, and lets the distribution fall above it with a tail temperature $T_t = r\,T_e$, continuous at the break. It needs r, and takes it from a table built from the model's own kinetic EEDF for a field-driven discharge: at each point, ground-state krypton at 300 K and 1 atm, as many ions as electrons, a field at the mean energy $\frac{3}{2}T_e$, electron-electron collisions at n_e/N and the growth term, as the run solves them; r is the tail ratio at which the two-temperature distribution removes the same energy through the 51 levels and the ionization as the kinetic one.

The excited axis. v0_4's table, $r(n_e/N, T_e)$, had no excited atoms in its states, and along v0_4's runs its tail lay below the kinetic EEDF's own. v0_5's is $r(n_e/N, T_e, x_s)$ on 1666 points: $\log_{10} n_e/N$ from -9 to -2.5, T_e from 0.5 to 8 eV, and the 5s fraction $x_s = [Kr^*]/[\text{Kr}]$ at 7 values from 0 to 1×10^{-3}. At each point the 5s levels hold x_s by their weights, and the 5p rung and Kr_2^* sit in their steady state against them at the point's T_e and n_e, from the model's own rates. Its $x_s = 0$ block is v0_4's table to the printed digit. The excited atoms' superelastic electrons heat the tail: at $n_e/N = 10^{-5}$ and 2.5 eV, r is 0.122 without them and 0.315 at $x_s = 10^{-3}$. Where the kinetic tail is colder than the form can follow (85 points) r takes the floor, 10^{-3}; where it is hotter (175 points, at low T_e with excited atoms, where the superelastic electrons put far more than the Maxwellian's loss into the tail), r takes the top, 3, the closest the form comes. Half and twice the pressure move r by at most 25 % at the points the runs visit, and 700 K gas by 16 %.

Along the reference run the table at the run's 5s fraction (7.1×10^{-5} to 1.0×10^{-4}) gives 0.96 to 0.99 of the kinetic EEDF's own tail ratio; its $x_s = 0$ block, v0_4's table, gives 1.16 to 1.30, and v0_3's beam table 1.84 to 1.99 (Table 7). Over the scan's corners the table's ratio is 0.95 to 1.07 of the kinetic one. The closure's 147-nm share then lies -8 % from the kinetic EEDF's at the reference point, where on the $x_s = 0$ block it lay +4 %: with the tail right, what is left is the Maxwellian below the break, where the stepwise ionizations' thresholds lie, and the zero block's hotter tail had more than offset it. At the scan's corners the closure is within 6 % of the kinetic EEDF, and v0_3's beam table, made for an electron beam, is +26 % off at the reference point.

		147 nm (% of E_{dep})			r over the kinetic's			
atm	kW cm^{-3}	kinetic	field table	v0_3's table	field	its $x_s = 0$	v0_3's	T_g end (K)
0.25	100	0.95	0.90 (−6)	0.89 (−7)	0.97	0.93	0.64	1053
0.25	500	0.01	0.00 (−6)	0.00 (−41)	0.95	0.92	0.48	3281
2	100	20.40	19.85 (−3)	27.93 (+37)	1.06	1.15	2.42	380
2	500	9.63	9.24 (−4)	12.90 (+34)	1.07	1.28	2.22	747
1	300	6.08	5.58 (−8)	7.67 (+26)	0.98	1.28	1.97	856

Table 7: The two-temperature closure against the kinetic EEDF at the scan's corners and the reference point: the 147-nm share on the kinetic EEDF and on the closure with the field table and with v0_3's beam table (the deviation in brackets), and the tables' tail ratio over the kinetic run's own, averaged over the pulse body (50 ns to the pulse's end): the field table at the run's x_s, its $x_s = 0$ block, and v0_3's.

5 The electrons and the excited atoms

v0_5 lets the electrons collide with every excited population: the four 5s levels, the ten levels of the 5p rung and both states of Kr_2^*. Under the kinetic EEDF each collision is a process of its own (74 in all), and on a solved EEDF each gives the electrons, or takes from them, its rate times its energy, a check every process passes.

Ionization of the excited atoms and of Kr_2^*. No measurement of it for krypton or argon was found. v0_5 takes Vriens and Smeets's semi-empirical cross section [69],

$$\sigma(E) = \frac{4\pi a_0^2 R^2}{E + 3.25\,E_n}\left(\frac{5}{3E_n} - \frac{1}{E} - \frac{2E_n}{3E^2}\right),$$

at each level's own ionization energy E_n: 4.08 eV for $1s_5$ down to 3.356 eV for $1s_2$, 2.70 eV for $2p_{10}$, and 3.64 eV for Kr_2^*, which it ionizes to Kr_2^+. Over a Maxwellian it gives their rate formula to within 8×10^{-3} for T_e of 1 to 5 eV (3 % at 0.5 eV). Its peak is 7.9×10^{-16} cm^2 at 11.1 eV for $1s_5$, 1.8×10^{-15} at 7.3 eV for $2p_{10}$ and 9.9×10^{-16} at 9.9 eV for Kr_2^*. Flannery et al.'s binary-encounter calculation gives Kr_2^* about 10^{-15} cm^2, 40 to 50 % above the metastable atom's [70]; the formula puts it 26 % above $1s_5$'s. Kannari et al. print fits of the same three rates [11], from Vriens's earlier calculation [13] and from Vriens and Smeets's: at the reference point's 1.57 eV they are 0.71, 1.02 and 0.79 times the formula (5s, 5p, Kr_2^*), and 0.62 to 1.27 times it from 1 to 3 eV. Each row's band runs from 0 to twice the formula.

At the reference point the electrons ionize 1.9 % of the 5s atoms formed, 0.9 % of the rung's and 56.9 % of the excimers formed (28.5 % radiate). In the pulse, on a Maxwellian at its mean T_e and n_e, the formula ionizes an excimer in about 74 ns, against its 270-ns radiative lifetime.

The superelastic return. Each 5s level returns to the ground state by the Klein–Rosseland reverse of its own excitation cross section, the electron gaining the level's energy. Blagoev et al. measured these rates for thermal electrons in a krypton afterglow [68]: 1.5×10^{-10}, 5.2×10^{-10} and 9.8×10^{-10} cm^{3}s^{-1} for $1s_5$, $1s_4$ and $1s_3$. For Maxwellian electrons at 300 to 2000 K the model's rates are 5.3 to 6.6, 5.5 to 9.5 and 0.5 to 1.2 times theirs (at 300 K, 8.8×10^{-10}, 5.0×10^{-9} and 1.2×10^{-9} cm^{3}s^{-1}). Such electrons read the excitation cross sections within a few tenths of an eV of their thresholds, where their shape is Biagi's: the training scales each group of cross sections as a whole. The measurement scores the returns and sets nothing. At the reference point they return 0.13 % of the 5s atoms formed and move the share by −0.5 %.

The 5s levels mixed by electrons. Blagoev and Pilosov measured $e^- + 1s_5 \rightarrow e^- + 1s_4$ at 4.3×10^{-8} cm^{3}s^{-1} in a krypton afterglow [71], the electron temperature not given in their abstract. v0_5 takes it as $A\,e^{-E_g/T_e}$ over the levels' gap E_g = 0.117 eV, with A their value (the band's low end, as if their electrons were hot against the gap) up to 2.3×10^{-7} (their value taken at 0.07 eV, the coldest of Qiu

et al.'s afterglow [72]), the reverse by detailed balance. At the reference point's mean n_e and T_e it turns $1s_5$ into $1s_4$ 1.44 times as fast as $1s_5$ forms the excimer, and removing it moves the share by -0.2 %.

The rows at 0. $e^- + Kr_2^* \rightarrow e^- + 2Kr$ and the mixing of the excimer's two states by electrons are neither measured nor calculated in the sources found. Their bands run from 0 to Kannari et al.'s estimate of the quenching, $10^{-7}\,cm^3s^{-1}$ [11]; under the kinetic EEDF each enters as a pseudo cross section whose Maxwellian rate is its rate law exactly.

The 5p rung's levels. The rung's ten levels are populated at the gas temperature: at 300 K 98 % of it is $2p_{10}$ (partition function 3.05), at 1000 K 51 %. A pump to a level above $2p_{10}$ pays its own threshold; the excess heats the gas as the levels relax, and the gas lends it back to a reverse, an ionization or a line out of that level. At 300 K the rung's lines give it a lifetime of 37.9 ns, nearly $2p_{10}$'s alone (v0_4: 29.0 ns, the levels' mean by weight). Its relaxation on krypton has its detailed-balance reverse, which reaches 1.8 % of it over the scan's run at 0.5 atm and 500 $kW\,cm^{-3}$, and 8×10^{-5} of it at the reference point's conditions (two-temperature closure).

6 The electron swarm, and what the training did

The kinetic EEDF's solver is run as a swarm: no electron-electron collisions, the temporal growth term on, and every grid sized so that the EEDF falls 11 to 25 decades across it. It gives the drift velocity (the Joule power over eE), the ionization coefficient $\alpha/N = k_i/W$ and the characteristic energy D_T/μ, with the transverse diffusion taken over the same momentum transfer and growth term as the field's flux [8]. Swarm measurements are measurements of the cross sections themselves, so they may train them, while the device and beam measurements of Table 1 may not. v0_3 trained a few factors on Biagi's level-resolved cross sections against the swarm data that agree with one another, and v0_4 and v0_5 run on them unchanged; this section is v0_3's, its numbers computed again on v0_5's code.

The data. 28 datasets, 441 points, from Dutton's survey [34] and LXCat's LAPLACE database [6, 7], which digitized them from the papers' tables or figures (Table 8). None prints an error, so each carries a class value: 3 % for a drift velocity measured after 1985, 5 % before, 7 % for a characteristic energy and 10 % for an ionization coefficient. A 2 % error in E/N enters through the model's own slope, which matters where the drift velocity climbs out of the Ramsauer minimum (0.01 to 0.03 Td) and for the ionization coefficient at low field. A dataset's χ^2 per point near 1 means the model agrees with it within these errors. Points above 300 Td are checks only: there the mean energy passes 7 eV, where the two-term approximation and the growth model part ways. The ionization coefficient below 50 Td never trains (below).

dataset	quantity	E/N (Td)	points	role	error (%)	χ^2 per point Biagi	χ^2 per point trained	rms (%) Biagi	rms (%) trained
Pack et al. 1962, 368 K [33, 34]	W	0.00153–0.00622	5	train	5	0.8	0.6	5	4
Hunter et al. 1988 [40]	W	0.002–3.0	42	train	3	0.8	0.7	3	3
Nakamura 1990 [42]	W	0.298–53	32	train	3	2.9	0.4	6	2
Pack et al. 1962, 300 K [33, 34]	W	0.000791–3.1	24	held out	5	4.5	4.3	12	11
Pack et al. 1992, 300 K [41]	W	0.0008–19	42	held out	3	14.2	17.7	15	15
Pack et al. 1962, 195 K [33, 34]	W	0.00121–0.0924	12	validation	5	3.1	2.8	10	9
Pack et al. 1992, 195 K [41]	W	0.0015–0.0894	10	validation	3	6.1	5.3	9	8
Bowe 1960 [44]	W	0.0247–7.4	53	held out	5	15.6	9.6	25	18
English and Hanna 1953 [45]	W	0.0399–2.0	31	held out	5	86.7	84.7	46	47
Dall'Armi et al. 1992 [43]	W	10–80	15	held out	3	10.3	13.4	11	12
Kucukarpaci and Lucas 1981 [46]	W	14–277	6	held out	5	39.6	48.4	25	26
the same, above 300 Td [46]	W	412–820	4	check	5	2.2	0.3	8	3
Koizumi et al. 1986 [47]	D_T/μ	0.005–0.9	26	validation	7	2.5	0.8	16	7
Al-Amin and Lucas 1987 [48]	D_T/μ	1.4–282	12	check	7	6.6	6.4	21	20
the same, above 300 Td [48]	D_T/μ	565–848	2	check	7	2.9	2.0	11	9
Kruithof 1940 [35, 34]	α/N	51–282	15	train	10	3.3	0.8	19	10
Heylen 1971 [36, 34]	α/N	55–250	21	train	10	0.9	0.2	10	5
Dall'Armi et al. 1992 [43]	α/N	55–80	6	train	10	1.8	0.8	15	11
Jacques et al. 1986 [5]	α/N	60–260	11	train	10	3.2	1.2	17	11
Bhattacharya 1979 [49]	α/N	62–281	11	train	10	0.4	1.3	8	15
Kruithof 1940, below 50 Td [35, 34]	α/N	14–45	11	held out	10	69.3	60.1	71	67
Specht et al. 1980, direct [50]	α/N	18–26	7	held out	10	15.0	10.4	53	46
Dall'Armi et al. 1992, below 50 Td [43]	α/N	20–50	7	held out	10	26.8	21.7	50	47
Heylen 1971, below 50 Td [36, 34]	α/N	28–49	10	held out	10	3.1	1.1	22	14
Jacques et al. 1986, 40 Td [5]	α/N	40–40	1	held out	10	0.0	1.2	2	17
Specht et al. 1980, total [50]	α/N	4.5–24	12	check	10	511	507	92	91
the same, above 300 Td [49]	α/N	312–625	4	check	10	1.1	2.7	12	19
the same, above 300 Td [35, 34]	α/N	339–1017	9	check	10	0.3	0.4	5	7

Table 8: Krypton's electron swarm data, their roles and the model's error on Biagi's file as published and on the trained set (v0_3's to v0_5's), each as the experiment measures it. Drift velocities W at 300 K unless labelled; Hunter et al.'s at 301 K. A validation set is held out of the training but took part in choosing which factors train (Table 10).

What a swarm experiment measures. The two-term solution with the temporal growth term gives the flux drift velocity and k_i/W. A drift tube times the swarm's centre of mass, the bulk drift velocity, and a steady-state Townsend experiment measures the spatial growth of its current, the SST coefficient. A Monte Carlo solution of the Boltzmann equation, with no two-term approximation, gives all of them [3]. Run on Biagi's file as published (Table 9), it finds the bulk drift velocity 23 % above the flux one at 300 Td, and the SST coefficient 9 % below the two-term's there. The two-term flux drift velocity matches the Monte Carlo's to 0.3 % from 10 to 300 Td, and from 40 Td up the two-term k_i/W lies 1.4 to 3.9 % below the Monte Carlo's flux coefficient. Below 20 Td, without ionization, the bulk and flux drift velocities are one, and the two-term drift velocity is compared as it is; its own error there (1.8 % at 0.2 Td) is absorbed by the trained factors, which are effective for the two-term solver. The two-term transverse diffusion is 46 to 79 % too high from 10 to 300 Td, so D_T/μ is scored only below 1 Td, and Al-Amin and Lucas's characteristic energies, 1.4 Td and up [48], are a check of the correction rather than of the cross sections. The model's two-term solution times these ratios is compared with each measurement. The ratios are taken on Biagi's file; on the trained set the Monte Carlo gives the drift velocity's and the ionization coefficient's to 1.6 %. Without them the model would be scored, and trained, against the wrong quantity: comparing the two-term coefficient with Townsend experiments puts it closer to them at high field than it is.

	Biagi's file			the trained set	
E/N (Td)	W_{bulk}	α_{SST}	D_T/μ	W_{bulk}	α_{SST}
20	0.999	1.036	0.63	–	–
30	1.003	1.036	0.60	1.002	1.034
40	1.006	1.036	0.58	–	–
50	1.011	1.025	0.57	1.018	1.008
75	1.038	0.997	0.56	–	–
100	1.065	0.979	0.56	1.070	0.973
150	1.113	0.956	0.59	–	–
200	1.152	0.939	0.62	1.165	0.930
300	1.232	0.906	0.68	1.241	0.908

Table 9: The Monte Carlo's value over the two-term solution's, LoKI-MC at 300 K and 1 Torr with 10^5 electrons [3]. The coefficient below 40 Td takes the 40-Td ratio, where the Monte Carlo counts too few ionizations to resolve it.

The training. The factors multiply Biagi's cross sections: the elastic momentum transfer by a factor read log-linearly between six energies (0.03 to 20 eV, held beyond them), and the excitation of the four 5s levels, the ten 5p levels and the 37 higher levels by one factor each. The factors at 0.03 and 0.2 eV stay at 1 (below), so 7 train. The ionization cross section is not trained: beam experiments measure it, and in a swarm its scale trades against the excitation's. The prior on each factor's logarithm is broad (standard deviation 0.5, a factor 1.65), so the data decide. Each trial of the factors needs the solver at every measured point, so the fits run on a neural network trained to reproduce it: 2000 sets of factors drawn over $\times 0.55$ to $\times 1.82$ (a Latin hypercube [52]), each solved at 81 values of E/N and gas temperature, and a three-layer network fitted to the change of each quantity from Biagi's file (Adam [54]). On designs it was not trained on it reproduces the solver to 0.10 % in the drift velocity, 0.13 % in D_T/μ and 0.74 % in the ionization coefficient, and every answer below is re-checked on the solver itself.

Which data train. The consistency check fits every candidate together, scores each against that joint fit, and drops the worst while its χ^2 per point exceeds 3. Of 16 candidates it drops 8: English and Hanna 1953 (73.7), Kucukarpaci and Lucas 1981 (44.6), Pack et al. 1992, 300 K (15.5), Dall'Armi et al. 1992 (16.4), Pack et al. 1992, 195 K (5.4), Pack et al. 1962, 300 K (5.0), Pack et al. 1962, 195 K (3.6), Bowe 1960 (3.5) (their χ^2 when dropped). The 8 that remain train. The choice of 3 matters little below it: at 2 the same sets remain. At 4, Pack et al. 1962, 195 K and Bowe 1960 join, and the 1960 drift velocities, -9 % from the 1988 ones where both exist on average, pull the momentum transfer at 6 eV to $\times 1.31$ and the 5s excitation to $\times 1.16$.

Which factors train. Two of the nine stay at 1, by a choice made on held-out data (Table 10). With all nine free the fit matches the train sets best (χ^2 0.49 per point pooled, against 0.64), but it lowers the momentum transfer at the Ramsauer minimum to $\times 0.88$ and raises it at 0.03 eV to $\times 1.13$, a trade the train sets allow and the data that weigh those energies differently reject: Koizumi et al.'s characteristic energy [47] goes from χ^2 2.5 per point on Biagi's file to 8.1, and Pack et al.'s drift velocities in gas at 195 K [33, 41] from 3.1 to 5.5 (1962) and from 6.1 to 8.7 (1992). With the factors at 0.03 and 0.2 eV held at 1, every train set stays within the consistency threshold (Hunter et al.'s drift velocity at 0.7 rather than 0.1), and those three fit better than on Biagi's file, at 0.8, 2.8 and 5.3, the best of the five variants. Holding the 0.6-eV factor too, or every momentum-transfer factor, fits both the train sets and these three worse. v0_3 trained that variant's 7 factors. Because these held-out sets chose it, they validate the training rather than test it. The tests that neither the training nor this choice saw are the ionization coefficient below 50 Td and the beam's partition (Section 8).

	train sets			held out (validation)		
factors trained	pooled	worst	Hunter 1988	Koizumi 1986	Pack 1962, 195 K	Pack 1992, 195 K
none (Biagi's file as published)	1.75	3.28	0.79	2.52	3.07	6.12
the three excitations	1.16	2.35	0.79	2.52	3.07	6.12
these, and the momentum transfer at 2 to 20 eV	0.66	1.26	0.75	2.43	2.89	5.29
these, and at 0.6 eV too (v0_3)	0.64	1.25	0.70	0.84	2.77	5.27
all nine	0.49	1.27	0.13	8.06	5.45	8.70

Table 10: Which factors train: each variant's least-squares factors on the train set, and the χ^2 per point on the solver itself, each dataset as its experiment measures it with Table 8's errors. Pooled is over the train sets' 143 points. v0_3 to v0_5 run on the fourth row's factors.

The fit. Least squares on the train set gives the factors; an ensemble MCMC [51] (acceptance 0.43) gives their posterior, and a neural posterior estimator, a network trained on simulated datasets to read the factors off them [53], gives the same answer independently (Table 11). Its $\pm 1\sigma$ intervals cover 0.60 to 0.65 of held-out simulations (ideal 0.68). The data pin the momentum transfer from 0.6 to 20 eV and the 5s and higher excitations (the network's R^2 at least 0.83); they barely pin the 5p excitation (R^2 0.15).

factor on	least squares	MCMC median	16–84 %	network	R^2
momentum transfer at 0.03 eV	1, not trained (Table 10)				
the same at 0.2 eV	1, not trained (Table 10)				
at 0.6 eV (the Ramsauer minimum)	1.157	1.157	[1.062, 1.260]	1.125 ± 0.09	0.93
at 2 eV	0.926	0.926	[0.891, 0.962]	0.909 ± 0.04	0.99
at 6 eV	1.039	1.040	[1.008, 1.071]	1.038 ± 0.03	0.99
at 20 eV and above	1.227	1.221	[1.106, 1.354]	1.240 ± 0.10	0.91
the four 5s levels (1s)	0.886	0.871	[0.776, 0.963]	0.838 ± 0.12	0.91
the ten 5p levels (2p)	0.925	0.954	[0.693, 1.293]	1.047 ± 0.34	0.15
the 37 higher levels	0.753	0.738	[0.631, 0.854]	0.703 ± 0.17	0.83

Table 11: The trained factors on Biagi's cross sections. v0_3 to v0_5 run on the least-squares values. The network column gives its mean and the standard deviation of the factor's logarithm; R^2 is the network's on held-out simulations.

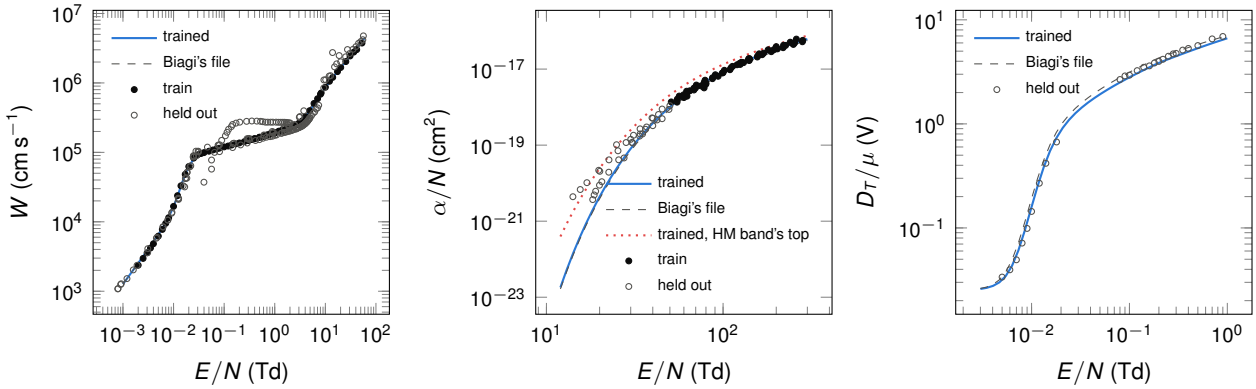


Figure 1: Krypton's drift velocity (left), first Townsend coefficient (middle) and characteristic energy (right) at 300 K, as the experiments measure them: the trained cross sections (solid), Biagi's file as published (dashed), and the measurements of Table 8, filled where they train and open where they are held out (the validation sets among them). Middle: the dotted curve is the trained set with every excitation of the levels at or above Kr_2^+'s energy ionizing associatively (the Hornbeck–Molnar band's top).

The scores.

- **The train sets.** The drift velocity of Hunter et al. [40], the most precise below 3 Td, moves from χ^2 0.8 to 0.7 per point, Nakamura's [42] from 2.9 to 0.4 (mean +5 to 0 %). The ionization coefficients of Jacques et al. [5] and Kruithof [35], which lay -15 and -18 % from the model, now lie -5 and -7 %; Bhattacharya's [49], which Biagi's file matched (-1 %), now lies +11 %. Every train set ends within the consistency threshold.
- **The held-out drift velocities** are the sets the check dropped. The training moves 5 of the 8 closer to the model, the two at 195 K among them, which took part in choosing which factors train. Three move further: Pack et al. 1992 at 300 K [41] (χ^2 14.2 to 17.7 per point), Dall'Armi et al.'s [43] (10.3 to 13.4) and Kucukarpaci and Lucas's [46] (39.6 to 48.4, -17 % from the model on average). Pack et al.'s 1992 set scatters by up to 30 % against the model between points within 5 % of each other in E/N; Bowe's [44] lie +16 % from the model, and English and Hanna's [45] -18 %.
- **The characteristic energy** of Koizumi et al. [47], held out, probes the momentum transfer near the Ramsauer minimum, which the drift velocities pin least: rms 15.8 % on Biagi's file and 7.1 % on the trained set. It chose which factors train, so it validates the training rather than tests it.
- **Below 50** Td, held out by rule, the mean error moves from -44 to -36 %: Heylen's from -20 to -9 %, Specht et al.'s direct coefficient [50] from -52 to -46 %, Kruithof's from -67 to -63 %.

Why the low field stays out. At 30 Td the electrons excite krypton 364 times for every ionization, so anything that ionizes an excited atom, or frees an electron at a surface with its light, can dominate the measured current. Specht et al. found the coefficient anomalously large at low field and attributed it to photoelectron emission at the surfaces by radiating metastables, which they say also explains why earlier measurements disagreed with transport calculations; their direct coefficient leaves that contribution out [50]. Bordage et al. found every published krypton set below the low-field data and ascribe it to Penning ionization of impurities [2]. And the 5 datasets part by up to $\times 4.0$ from one another where they overlap. Training on them would fit the cross sections to the purity and geometry of each experiment. Pure krypton has one process there of its own, the associative (Hornbeck–Molnar) ionization of atoms excited above 12.85 eV [4]. At the band's top it raises the trained set's coefficient $\times 12.7$ at 14 Td, $\times 3.1$ at 30 Td and $\times 1.5$ at 100 Td, and puts the model +54 % from the train sets on average, so a share that large cannot hold where those are measured. No krypton measurement gives the share, and it stays at 0 by default.

What this means for the reference point. The electrons sit at 1.34 to 6.4 Td. There direct ionization makes 0.089 % of the ions, and the ionization of the excited atoms and of Kr_2^* with Penning ionization the rest, so the low-field coefficient does not reach the answer. The excitation rates do, and the training moves them: on Biagi's file as published v0_5 gives 6.19 % at the reference point, and on the trained set 6.08 % (-1.8 %); the posterior of the factors spreads it by ± 0.06 points (Section 1).

7 The electroluminescence yield

Mano et al. measured the secondary scintillation of pure krypton at 1.1 bar and room temperature: the photons an electron drifting in a uniform field makes per centimetre, over the gas density [66]. Below the field at which the electrons ionize, every photon comes from an excitation, so the yield measures the excitation cross sections in the swarm regime, where no electron meets another and no excited atom meets an electron. Their fit, $Y/N = 0.113\,E/N - 0.3$ in 10^{-17} photons per electron per atom per cm^{-2} (E/N in Td), holds from its intercept at 2.7 Td to 14 Td. Nothing in v0_5 or in its training saw it.

v0_5's swarm (Section 6) gives the excitations per electron and per centimetre of drift, $\sum k_{\mathrm{exc}}/W$, level by level. Each excitation is then weighted by the photons the model's own network makes of it at 1.1 bar and 296 K: a dilute seed of each 5s level and of the 5p rung, with no drive and no electrons to speak of, run to the end of its cascade. At the bands' defaults every seed ends as one 147-nm photon (1.000 per $1s_5$ atom, 1.000 per $1s_4$ atom, 1.000 per 5p atom); with Kr_2^* + Kr at its band's top, 0.941 per 5p atom.

	Y/N (10^{-17} cm^2)		model over fit		
E/N (Td)	Mano et al.'s fit	model	bands' defaults	Kr_2^* + Kr at its top	5s share (%)
4	0.152	0.099	0.65	0.61	100.0
5	0.265	0.237	0.89	0.84	100.0
6	0.378	0.375	0.99	0.93	99.9
8	0.604	0.628	1.04	0.98	99.7
10	0.830	0.857	1.03	0.97	99.2
12	1.056	1.075	1.02	0.96	98.5
13.5	1.226	1.233	1.01	0.95	97.8

Table 12: The electroluminescence yield of pure krypton: Mano et al.'s fit [66] and v0_5's swarm at 296 K, each excitation weighted by its photons from the network's cascade at 1.1 bar; the 5s share is the excitations into the four 5s levels.

From 6 to 13.5 Td the model lies at 0.99 to 1.04 of the fit (2.4 % rms), and with the excimer's quenching at its band's top at 0.93 to 0.98. Toward the fit's intercept it falls below it (0.89 at 5 Td, 0.65 at 4): there the fit's yield is 0.22 and 0.12 of its value at 13.5 Td, and a straight line's intercept is the least constrained part of a fit to a yield that bends into its threshold. Nearly every excitation below 14 Td is a 5s level's, so the score tests the four 5s cross sections and the trained 5s factor ($\times$0.89) at the swarm's mean energies of 4.3 to 4.5 eV, and the excitation-to-photon chain at a density where nothing competes with the light.

8 Under an electron beam

Three measurements bear on the chain from excitation to the 147-nm photon. They were made under electron beams, so for them v0_5 runs under a beam drive. The drive is a fixture of this note, not part of the model. It deposits the given power as ion pairs, 5s and 5p atoms and secondary electrons in the proportions of a fast electron's degradation in pure krypton. That degradation is the Fowler–Spencer–Fano recursion on the same cross sections as the model, level by level, from 528 eV: an excitation continues the electron at $E - I$, and an ionization shares $E - I$ equally between the two electrons. Per ion pair it gives:

- W = 26.2 eV;
- 0.410 Kr**, from the levels above the 5s manifold;
- 0.273 5s atoms: 0.045 $1s_5$, 0.134 $1s_4$, 0.007 $1s_3$ and 0.088 $1s_2$, the two resonance levels most, as a fast electron's dipole excitation favours them (v0_3 put the four into one Kr*);
- 0.46 eV of heat, the higher levels' excess over the 5p rung's energy;
- 4.29 eV of sub-excitation secondaries, which the kinetic equation injects with a flat spectrum up to 9.9152 eV.

That is one excitation or ion per 15.55 eV.

The partition. Saito et al. [37] measured scintillation photons and electrons together under 5.49-MeV alpha particles in 1 to 10 atm of krypton. They found 0.55 excited atoms per ion pair, and W_s = 15.4 eV, the energy per photon when every pair recombines. They also confirmed that each recombination gives one photon. On the trained cross sections the drive's 15.55 eV per excitation or ion is +1.0 % from their W_s, its excitations per pair (0.68) differ from theirs by +24 %, and its W differs by +10 % from the 23.9 eV their two numbers imply. On Biagi's file as published the same degradation gives W = 27.6 eV (+16 %) and 0.79 excitations per pair (+44 %): the swarm data's smaller excitation moves both toward the measurement, which neither the training nor the choice of its factors saw. One photon of 8.43 eV per 15.4 eV is an efficiency of 0.547. The model's ceiling, one photon per excitation and per ion of the drive, is 0.542.

Low pump. Morozov et al. [38] measured the power a continuous 8 to 15-keV beam deposits in the gas against the VUV power emitted into 4π. For krypton they found 0.42 $\pm$ 0.05, flat from 500 to 1400 hPa

and over a tenfold range of beam power. At 1000 hPa, 30 and 300 W cm^{-3} for 1 μs with the light counted for 50 μs after, v0_5 gives 0.542 and 0.543 (Table 13). The error is +29 %, 2.5 of the measurement's standard errors (v0_4: +29 %). At this pump nothing competes with the light: 0.001 Penning ionizations per pair, every ion recombined (1.010 per pair), 0.000 excimers redissociated and 0.005 ionized per pair, so nearly every Kr_2^* radiates. So v0_5 sits at its ceiling, which Saito et al.'s measurement supports. Morozov et al. compare their value with calculated maxima of 0.46 and 0.52 for krypton. The 23 % between their value and Saito et al.'s one photon per excitation and per ion (0.547) is not explained in v0_5.

High pump. Eckstrom et al. [39] report the fluorescence efficiency of Kr_2^* near their theoretical limit of about 46 %. They found it under a low-current, long-pulse beam at modest pressures, for pump rates up to 3×10^5 W cm^{-3}. The abstract does not give the pressure of each point. At that pump rate (1 μs, the light counted for 20 μs) v0_5 gives:

- 0.093 at 0.5 atm, 0.445 at 1 atm and 0.535 at 2 atm, that is 83, 18 and 1 % below its ceiling of 0.542 (v0_4: 0.131, 0.466 and 0.543);
- at 1 atm, Penning ionization of the 5s levels on one another costs 0.270 photons per pair, because the excimer forms too slowly to outrun it entirely;
- at 1 atm with Kr* + Kr* at its band's low end, 0.497 (8 % below the ceiling), and with the $1s_5$ formation at its band's top, 0.450 (17 %).

The rows v0_5 adds act under the beam only where its plasma is dense. At low pump the efficiency is v0_4's, 0.543. At 3×10^5 W cm^{-3} and 1 atm the electrons ionize 0.137 excimers per ion pair and the stepwise ionizations make 0.043 more ions, and the light changes by -5 % against v0_4; at 0.5 atm, where the gas passes 1364 K, by -29 %, and at 2 atm by -2 %.

run	measured	kinetic	two-temperature	v0_4, kinetic	Penning per pair	photons per pair	T_e (eV)	T_g end (K)
30 W cm^{-3}, 1000 hPa	0.42 ± 0.05	0.542	0.541		0.000	1.681	0.75	300
300 W cm^{-3}, 1000 hPa	0.42 ± 0.05	0.543	0.541	0.543	0.001	1.686	0.83	300
3×10^5 W cm^{-3}, 0.5 atm	near 0.46	0.093	0.096	0.131	0.700	0.289	0.42	1364
3×10^5 W cm^{-3}, 1 atm	near 0.46	0.445	0.446	0.466	0.270	1.381	0.48	628
3×10^5 W cm^{-3}, 2 atm	near 0.46	0.535	0.532	0.543	0.032	1.660	0.64	438
1 atm, Kr* + Kr* 1.5×10^{-10}		0.497	0.491		0.105	1.544	0.45	597
1 atm, $1s_5$ + 2Kr 4.2×10^{-32}		0.450	0.451		0.254	1.398	0.48	624

Table 13: v0_5 under the beam drive: the VUV efficiency (147-nm energy over the deposited energy) on both closures and v0_4's on the kinetic EEDF, and on the kinetic EEDF the Penning ionizations and 147-nm photons per ion pair, the pulse's mean T_e and the final gas temperature. The ceiling, one photon per excitation and per ion, is 1.683 photons per pair, an efficiency of 0.542; Saito et al.'s ideal is 0.547.

9 The energy ladder

Figure 2 draws v0_5's levels at their formation energies and the rows that move energy between them.

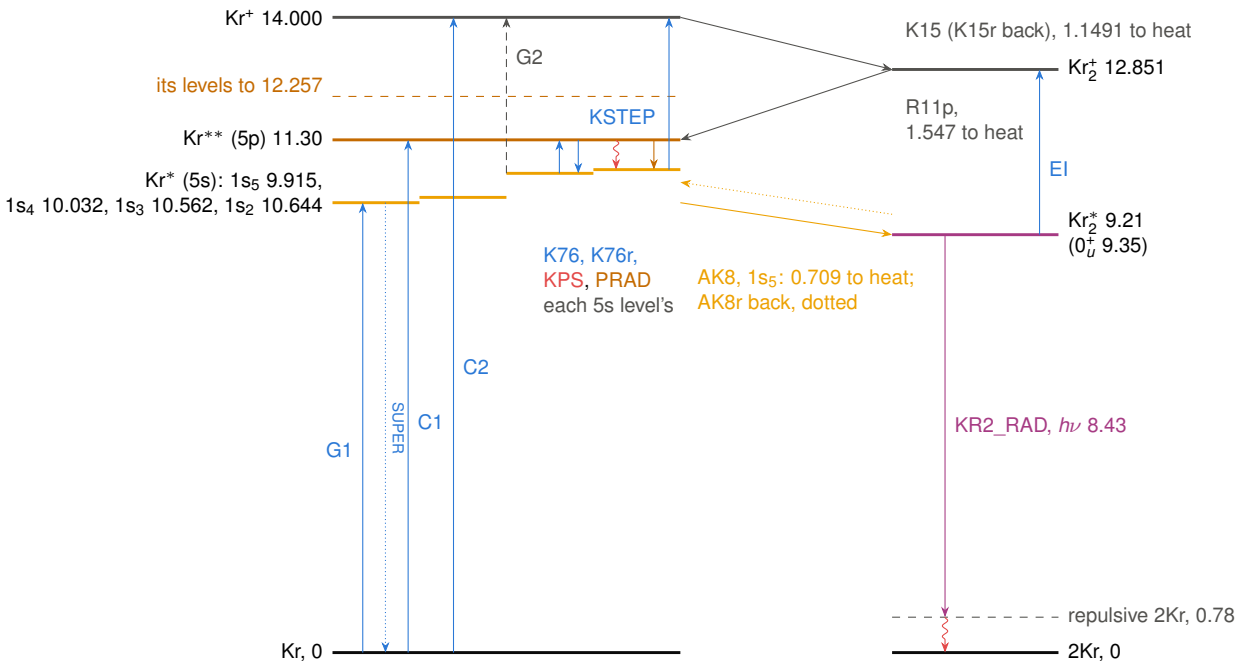


Figure 2: v0_5's energy ladder in eV above ground-state krypton at rest. A bar is a species at its formation energy; the four 5s levels are drawn side by side. Blue arrows are electron impact, which takes its energy from the electron gas; K76r, the 5p $\rightarrow$ 5s arrow beside K76, returns it; v0_5's KSTEP (KSTEP_S and KSTEP_P) ionize the 5s levels and the rung (3.356 to 4.084 eV and 2.696 eV from $2p_{10}$), EI the excimer (3.644 eV), and SUPER (dotted) returns a 5s level to the ground state, its energy to the electron. The dashed bar is the rung's top level; its levels are populated at the gas temperature, and a pump to one above $2p_{10}$ heats the gas with the excess as they relax. The dashed G2 is the Penning ionization of the 5s levels on one another, which gives the electron 5.831 eV for two $1s_5$ atoms. A red squiggle is heat: KPS relaxes the 5p rung on krypton, and PRAD radiates its gap to each level, 0.660 to 1.388 eV, in the near infrared. The excimer forms out of each 5s level, its well and the level's excess to heat, redissociates by detailed balance (AK8r), and radiates onto the repulsive ground state, which falls apart and heats the gas. The 5s levels exchange population on krypton (MIX, not drawn). C1's levels, 11.30 to 13.60 eV, cascade to the 5p rung, each heating the gas with its excess. HM, the Hornbeck–Molnar ionization of the levels above Kr_2^+'s energy, and the resonance lines of $1s_4$ and $1s_2$ are bands at 0 and not drawn.

10 Where v0_5 is not yet reliable

- **The excimer's formation.** Its rates still decide much of the answer. The $1s_5$ rate's measurements spread by a factor of 2.5; the $1s_4$ rate is one measurement [55], which Audouard et al.'s formation-time ratio puts three times lower [56] and Janssens et al.'s quasiclassical calculation 1.5 times higher [65]; the $1s_3$ rate is a proxy, Kholin et al.'s measurement having argon for its third body [67]. The temperature law is measured over 200 to 400 K [24], and the reference point ends at 856 K.
- **The calculated rows.** The stepwise ionizations of the 5s levels and of the 5p rung and the ionization of Kr_2^* by electrons run on Vriens and Smeets's calculation [69], and no krypton measurement tests it. At the reference point they make 33.8, 16.0 and 45.2 % of the ions, and their bands, 0 to twice the calculation, take the 147-nm share from 6.96 to 5.60 % (5s), 6.37 to 5.83 % (5p) and 15.43 to 4.47 % (Kr_2^*). Kannari et al.'s fits of the same rows [11] lie 0.62 to 1.27 times the calculation for T_e of 1 to 3 eV, and the calculation's peak for Kr_2^*, 9.9×10^{-16} cm^2, is the order of Flannery et al.'s binary-encounter cross sections [70].
- **The electrons' other collisions with Kr_2^***, its quenching to the ground state and the mixing of its two states, are not measured; the one value found is Kannari et al.'s estimate of the quenching, the top

of both bands [11]. Both are 0 by default; at the top the quenching takes the share to 1.27 % and the mixing to 12.44 %.

- **The superelastic returns** of the 5s levels are the detailed-balance reverses of Biagi's excitations. For thermal electrons the model's rates are 5.3 to 6.6 times Blagoev et al.'s [68] for $1s_5$, 5.5 to 9.5 times for $1s_4$ and 0.5 to 1.2 times for $1s_3$ (Section 5); such electrons read the excitations within a few tenths of an eV of their thresholds. At the reference point the returns move the share by -0.5 %.
- **The 5s levels' mixing by electrons** carries $1s_5 \leftrightarrow 1s_4$ alone, on Blagoev and Pilosov's afterglow rate [71], measured at an electron temperature their abstract does not give; the band's top takes it as measured at 0.07 eV. Qiu et al. measured the $1s_5 \rightarrow 1s_4$ and $1s_3 \rightarrow 1s_2$ transfers for T_e of 0.07 to 1 eV [72], values not in a source in hand; the other pairs are not carried.
- **The 5p rung's levels** are populated at the gas temperature, as if collisions with krypton mixed them faster than they radiate, are pumped or relax. That sets which level a pump's reverse, a stepwise ionization and a line leave from. Whitehead et al. measured the deactivation of the two-photon-excited 5p levels in krypton [74], values not in a source in hand.
- **The field-driven regime is unscored.** No pure-krypton measurement in hand tests the 5s $\leftrightarrow$ 5p cycle under a field. Its cross sections are measured to $\pm$35 to 40 % out of the two metastables; those of the two $J = 1$ levels are not measured and take their metastable's (band `k_5s5p_j1`). Zeng et al.'s relativistic R-matrix calculation gives them [73], its values not in a source in hand.
- **The redissociation** rests on the excimer's well, whose measurements give 0.62 to 0.709 eV (its band), and on a rotational constant from an estimated bond length, 2.6 Å, the ion's, which enters as $1/r_e^2$; the ion's vibrational quantum is taken as the excimer's. At the reference point 14.0 % of the excimers formed redissociate; above 988 K at the fill's density most would.
- **Hot gas.** v0_5 carries the rung's reverse, which reaches 1.8 % of its relaxation at the scan's 0.5 atm and 500 kW cm^{-3}. The scan's 0.1-atm column ends at 1354 to 3064 K, outside the measured ranges of the formation and the recombination.
- **The excimer's two states.** The fast 0_u^+ state takes no share by default: its measured share is above 6×10^{19} cm^{-3} and appears from 1.5×10^{19} [56], so at 1 atm it lies between 0 and the band's top. Quenching by krypton is 0 by default, against one measurement of a pressure-dependent decay at 984 nm [63] that may be a transfer between radiating states rather than a loss. No rate mixes the two states on krypton, and the electrons' mixing is a band at 0.
- **The two-temperature closure** now reads its tail at the run's excited fraction. With that axis it lies -8 % from the kinetic EEDF at the reference point (on v0_4's table, without the axis, +4 %) and within 6 % at the scan's corners. Its tail now follows the kinetic EEDF's along the reference run (0.96 to 0.99 of it), so what is left is its Maxwellian below the break, which v0_4's table had more than offset. The default closure does not use it.
- **The trained cross sections** rest on errors the swarm data do not print (class values here) and on the consistency check's threshold (at 4 rather than 3, two more sets train and the momentum transfer at 6 eV moves to $\times$1.31). The data barely pin the 5p excitation. Which factors train was chosen on held-out data (Table 10), so those sets validate the training rather than test it. Their posterior is propagated into the answer ($\pm$0.06 points).
- **The rows with no measurement or calculation.** Kr_2^* + Kr_2^*, the electrons' quenching and mixing of Kr_2^* and the Hornbeck–Molnar share are 0 by default (Table 2). At its band's top Kr_2^* + Kr_2^* moves the 147-nm share by -16 %. The $J = 1$ levels' 5s $\rightarrow$ 5p cross sections are proxies. With the excimer's ionization, the two rows on Kr_2^* at 0 make the uncertainty's three largest terms (Section 1).
- **Kr* + Kr*.** Its 5.0×10^{-10} cm^{3}s^{-1} is a 1974 measurement [18] as Kannari et al. print it [11], taken for every pair of levels.
- **The e-beam efficiency at low pump** is off Morozov et al.'s by +29 %, and v0_5 holds no process that would explain the difference.
- **The ionization coefficient below 50** Td lies -36 % from the measurements on average, which part by up to $\times$4.0 from one another; every published cross-section set falls below them [2]. It is held out of the training by rule, and it does not reach the reference point, where the ionization of the excited atoms and of Kr_2^* makes the ions.
- **The Hornbeck–Molnar share** of the levels above 12.85 eV is not measured for krypton and is 0

by default. At its top it puts the model +54 % from the train sets and moves the reference point by -0.01 %.

- **Processes v0_5 does not carry**, each needing a measured rate or a calculation before it can enter:
  - Kr_3^+ and its recombination;
  - Penning ionization between Kr^* and Kr_2^*, and of Kr^{**};
  - the electrons' excitation of the 5s and 5p levels above the rung, their transfers between the 5s levels other than $1s_5 \leftrightarrow 1s_4$, and their return of the rung to the ground state.

  Three-body recombination $Kr^+ + e^- + e^-$ [32] is not carried either: at 1 atm Kr^+ converts to Kr_2^+ in 7 ns.

How the numbers were made. Every number in this note is computed from the model when the note is built. None is typed by hand except the measurements, which carry their sources, and the training's record, which the model carries beside its cross sections and the training wrote. The runs are cached under a hash of the model's code, so a change to the model reruns them all (code `b392d33a7c61`). The kinetic runs take about 21 minutes each; the two-temperature runs take seconds.

A Reaction rates

The rows of v0_5 with the values they run with and their sources, a <level> or <1s> row once per 5s level and a <2p> row once per level of the 5p rung. Electron-impact rows use their cross sections over the kinetic EEDF (or the two-temperature closure). A band gives its default and its two ends.

row	reaction	value	source
G1_<level>	$e^- + Kr \to e^- +$ Kr*(level)	each 5s level at its threshold, 9.9152 to 10.644 eV, its own species	Biagi, Magboltz 8.97 [1, 6, 7], $\times 0.89$ (trained, Table 11)
C1	$e^- + Kr \to e^- +$ Kr**	the 5p, 4d, 6s and higher levels, 11.30 to 13.60 eV, $\times 0.92$ (5p) and $\times 0.75$ (higher) trained; the excess over 11.30 to heat	Biagi, Magboltz 8.97
HM	$e^- + Kr (+ Kr) \to Kr_2^+ + 2e^-$	the share of the excitations of the 20 levels at or above 12.85 eV; band 0 (default) to 1	no measurement for krypton; the process, Hornbeck and Molnar [4]
C2	$e^- + Kr \to Kr^+ + 2e^-$	13.9996 eV	Biagi, Magboltz 8.97
G6	e^- + Kr, elastic	the momentum transfer cross section, times the trained factor	Biagi, Magboltz 8.97
K76_<1s>_<2p>, K76r	$e^- +$ Kr*(1s) $\rightleftarrows e^- +$ Kr**(2p)	the measured cross section of each of the 26 pairs out of $1s_5$ and $1s_3$, the $J = 1$ levels on their same-core metastable's, each at its own threshold on Biagi's level energies, the level's excess over $2p_{10}$ to heat; band $\times$ 0.6 to 1.4, and $\times$ 0 to 1 on the $J = 1$ levels; the reverse by detailed balance out of the rung's level	Jung et al. [10, 9]; Kannari's fit as a comparison [11, 12]
KSTEP_S_<level>, KSTEP_P_<2p>	$e^- +$ Kr*(level), Kr**(2p) $\to Kr^+ + 2e^-$	Vriens and Smeets's cross section at the level's ionization energy, 4.08 eV out of $1s_5$ and 2.70 eV out of $2p_{10}$, the rung's levels at T_g; bands $\times$ 1, 0 to 2	calculation [69], no measurement; Kannari et al.'s fits of Vriens's calculation [11, 13] as a comparison
EI_KR2, EI_KR2F	$e^- + Kr_2^*$, $Kr_2^*(0_u^+) \to Kr_2^+ + 2e^-$	Vriens and Smeets's cross section at the excimer's ionization energy, 3.64 eV; band $\times$ 1, 0 to 2	calculation [69]; Flannery et al.'s order [70]; Kannari et al.'s fit [11]
EQ_KR2	$e^- + Kr_2^* \to e^- + 2Kr$	0; band to 1.0×10^{-7} cm^{3}s^{-1}; the electron gains the 147-nm quantum	no measurement; Kannari et al.'s estimate is the top [11]
EM_KR2, EM_KR2r	$e^- + Kr_2^* \rightleftarrows e^- + Kr_2^*(0_u^+)$	0; band to 1.0×10^{-7} cm^{3}s^{-1} on e^{-E_g/T_e}, E_g the states' gap, 0.142 eV; the reverse by detailed balance	no measurement
SUPER_<level>	$e^- +$ Kr*(level) $\to e^- +$ Kr	G1's cross section reversed by detailed balance; the electron gains the level's energy	Biagi, Magboltz 8.97, trained; scored on Blagoev et al.'s thermal rates [68]
EMIX_1s5, EMIXr_1s4	$e^- +$ Kr*($1s_5$) $\rightleftarrows e^- +$ Kr*($1s_4$)	$4.3 \times 10^{-8}\, e^{-E_g/T_e}$ cm^{3}s^{-1}, E_g the levels' gap, 0.117 eV; band to 2.3×10^{-7}, the measurement taken at 0.07 eV; the reverse by detailed balance	Blagoev and Pilosov [71]; Qiu et al.'s range of T_e [72]
R11p	$Kr_2^+ + e^- \to$ Kr** + Kr	$1.6 \times 10^{-6}\,(300\,\mathrm{K}/T_e)^{0.55}$ cm^{3}s^{-1}, ± 10 %; a gas-temperature factor, band 0 to 1	Shiu and Biondi [30]; Mikus within 8 % [31]; Cunningham and Hobson [62]

Table 14: v0_5's electron collisions. Rates in cm^3 s^{-1}, T_e in eV.

row	reaction	value	source
KPS_<level>, KPSr	Kr^{**} + Kr ⇄ Kr^*(level) + Kr	1.7×10^{-10} $cm^3 s^{-1}$ in all, over the levels by $2J + 1$ (band `kps_5s`: 1, to all into $1s_5$); band 3.0×10^{-11} to the default; the reverse by detailed balance on the rung's levels at T_g	Gazeli et al. [14], one level, products unresolved; the low end, the range Chang and Setser measured for argon's 4p levels [16]
PRAD_<2p>_<1s>	$Kr^{**}(2p) \rightarrow Kr^*(1s) + h\nu$, near IR	NIST's Einstein coefficient of each of the 28 lines, out of the rung's levels at T_g: at 300 K the rung's lifetime is 37.9 ns, nearly all $2p_{10}$'s (v0_4: 29.0 ns, the $2J + 1$ mean)	NIST ASD [75]; the measured lifetimes [17] and a TALIF lifetime of one level, 34.1 ns [15], as cross-checks
G2_<a>_<b>	Kr^*(a) + Kr^*(b) → Kr^+ + Kr + e^-	5.0×10^{-10} $cm^3 s^{-1}$ for each pair (twice for two levels); band 1.5×10^{-10} to the default	Leichner and Ericson [18] as Kannari et al. print it [11]
AK8_1s5	$Kr^*(1s_5)$ + 2Kr → Kr_2^* + Kr	3.8×10^{-32} $cm^6 s^{-1} \times (T_g/300)^{-0.64}$; band 1.7×10^{-32} to 4.2×10^{-32}	Marchal et al. [55]; Ueda et al. [19], Tracy and Oskam [21], Turner [58] and Smith and Turner [22] span the band; the law, Wieme and Lenaerts, 200 to 400 K [24]
AK8_1s4, AK8f	$Kr^*(1s_4)$ + 2Kr → Kr_2^* + Kr	9.2×10^{-32} $cm^6 s^{-1}$, the same law; band to a third of the $1s_5$ rate; the share into the fast 0_u^+ state, band 0 to 0.17	Marchal et al. [55]; Audouard et al. [56]
AK8_1s3	$Kr^*(1s_3)$ + 2Kr → Kr_2^* + Kr	the $1s_5$ rate as a proxy; band 0 to the $1s_4$ rate	Kholin et al.'s measurement has argon for its third body [67]
AK8r_<level>	Kr_2^* + Kr → Kr^*(level) + 2Kr	detailed balance on the excimer's well, D_0 0.709 eV; band to 0.62 eV	LaRocque et al. [59], Killeen and Eden [60]
MIX	$Kr^*(1s_4)$ + Kr → $Kr^*(1s_5)$ + Kr, and $Kr^*(1s_2)$ + 2Kr → $Kr^*(1s_4)$ + 2Kr	1.3×10^{-13} $cm^3 s^{-1}$ at 295 K on $(T_g/295)^{-1.29}$ through Turner's 196-K point, and 0.2×10^{-32} $cm^6 s^{-1}$; their reverses by detailed balance	Turner [58]; Audouard et al. [57]
RES	$Kr^*(1s_4, 1s_2) \rightarrow$ Kr + $h\nu$, 123.6 and 116.5 nm	the escaping share; band 0 to 1.7×10^5 s^{-1}	the pressure-independent term of Janssens et al.'s loss rate [65]
KR2_RAD	$Kr_2^* \rightarrow$ 2Kr + $h\nu$, 147 nm	τ = 270 ns, $h\nu$ = 8.43 eV	Morikawa et al., 270 ± 5 ns [25]; 250 ± 20 [20], 280 ± 30 [23]
KR2F_RAD	$Kr_2^*(0_u^+) \rightarrow$ 2Kr + $h\nu$	τ = 3.4 ns	Morikawa et al. [25]
QKR2	Kr_2^* + Kr → 3Kr	0; band to 8.6×10^{-15} $cm^3 s^{-1}$	Oka et al. [64]; Firestone and Selander [63]
PEN_KR2	Kr_2^* + Kr_2^* → Kr_2^+ + 2Kr + e^-	0; band to 5.0×10^{-10}	no measurement; Kannari et al.'s value is the top [11]
K15, K15r	Kr^+ + 2Kr ⇄ Kr_2^+ + Kr	2.4×10^{-31} $cm^6 s^{-1} \times (T_g/300)^{-0.75}$; the reverse by detailed balance on the ion's 1.1491 eV	the mean of three measurements [26, 27, 28]; the law, Pérez-Ríos and Greene [29]; Signorell et al. [61]

Table 15: v0_5's heavy-particle rows and radiation. Rates in $cm^3\,s^{-1}$ or $cm^6\,s^{-1}$. Kr_2^+ lies 1.1491 eV below Kr^+ + Kr [61].

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